

Experiment 7: Random Sampling and Sampling Distribution

```
import numpy as np
import matplotlib.pyplot as plt
population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(
    population_mean,
    population_std,
    population_size
)
sample_sizes = [30, 50, 100]
num_samples = 1000
sample_means = {}
for size in sample_sizes:
    sample_means[size] = []
    for _ in range(num_samples):
        sample = np.random.choice(
            population,
            size=size,
            replace=False
        )
        sample_means[size].append(np.mean(sample))
plt.figure(figsize=(12, 8))
for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i + 1)
    plt.hist(
        sample_means[size],
```

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bins=30,
alpha=0.7,
label=f'Sample Size {size}'
)
plt.axvline(
    np.mean(population),
    color='red',
    linestyle='dashed',
    linewidth=1.5,
    label='Population Mean'
)
plt.title(f'Sampling Distribution (Sample Size {size})')
plt.xlabel('Sample Mean')
plt.ylabel('Frequency')
plt.legend()
plt.tight_layout()
plt.show()

```

Output:



